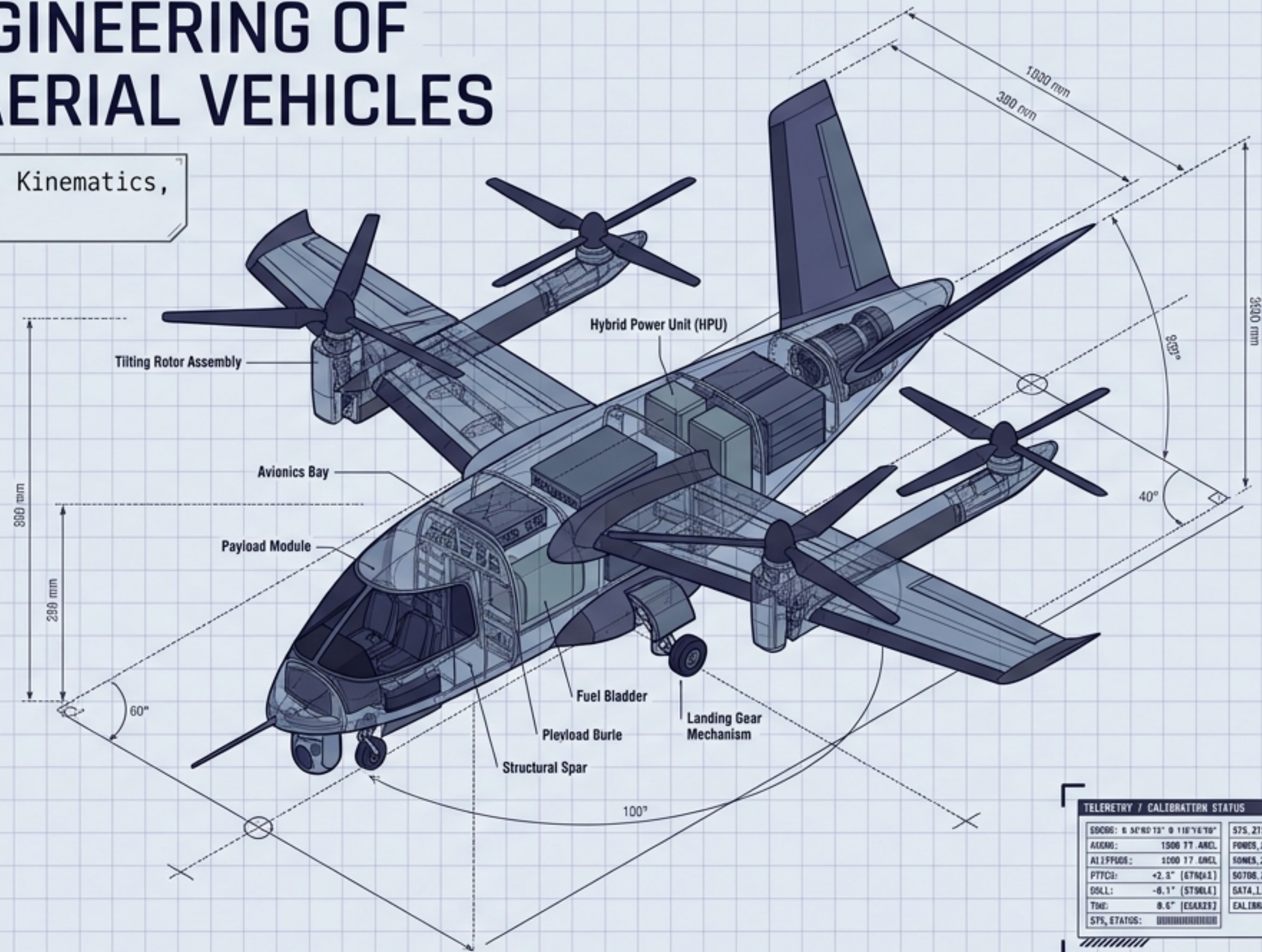


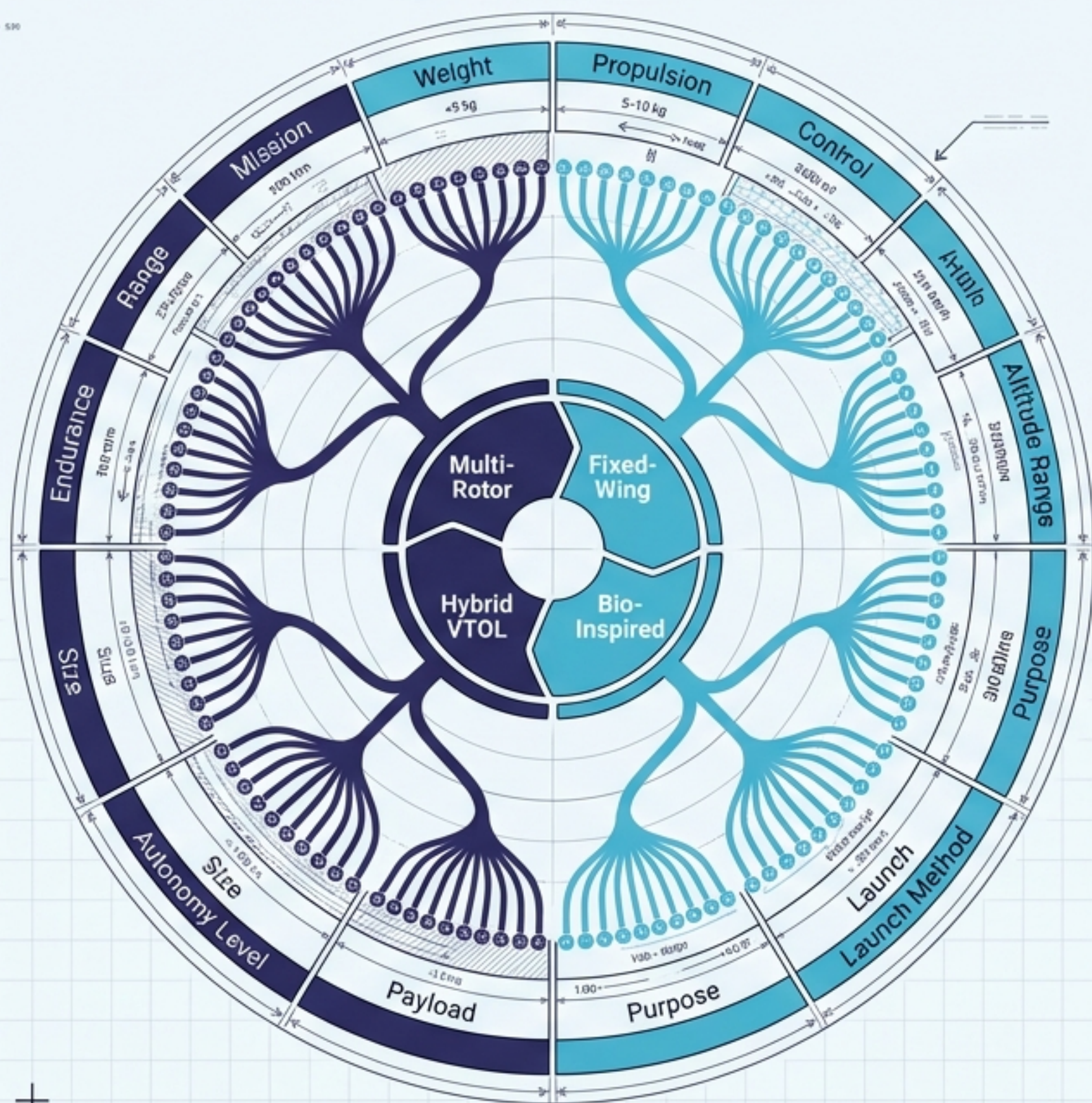
SYSTEMS ENGINEERING OF UNMANNED AERIAL VEHICLES

Aero-Structural Diagnostics, Kinematics, and Configuration Physics



TELEMETRY / CALIBRATION STATUS			
SOURCE: 6 50ND13 0 110V470	STATUS: ██████████	POWER: 200	STATUS: 300AM 130
ALIFFUG: 1000 17.000L	STATUS: 200	STATUS: 200	20.40 / 800
PTTC: +2.3" (6700L)	STATUS: 200	STATUS: 200	4000 (16L)
OSL: -8.1" (5700L)	STATUS: 200	STATUS: 200	STATUS: 100%
TOW: 8.6" (5000L)	STATUS: 200	STATUS: 200	STATUS: COMPLETE (OK)
STS, STATUS: ██████████			

Macro-Level Classification Taxonomy and NATO Sizing Framework



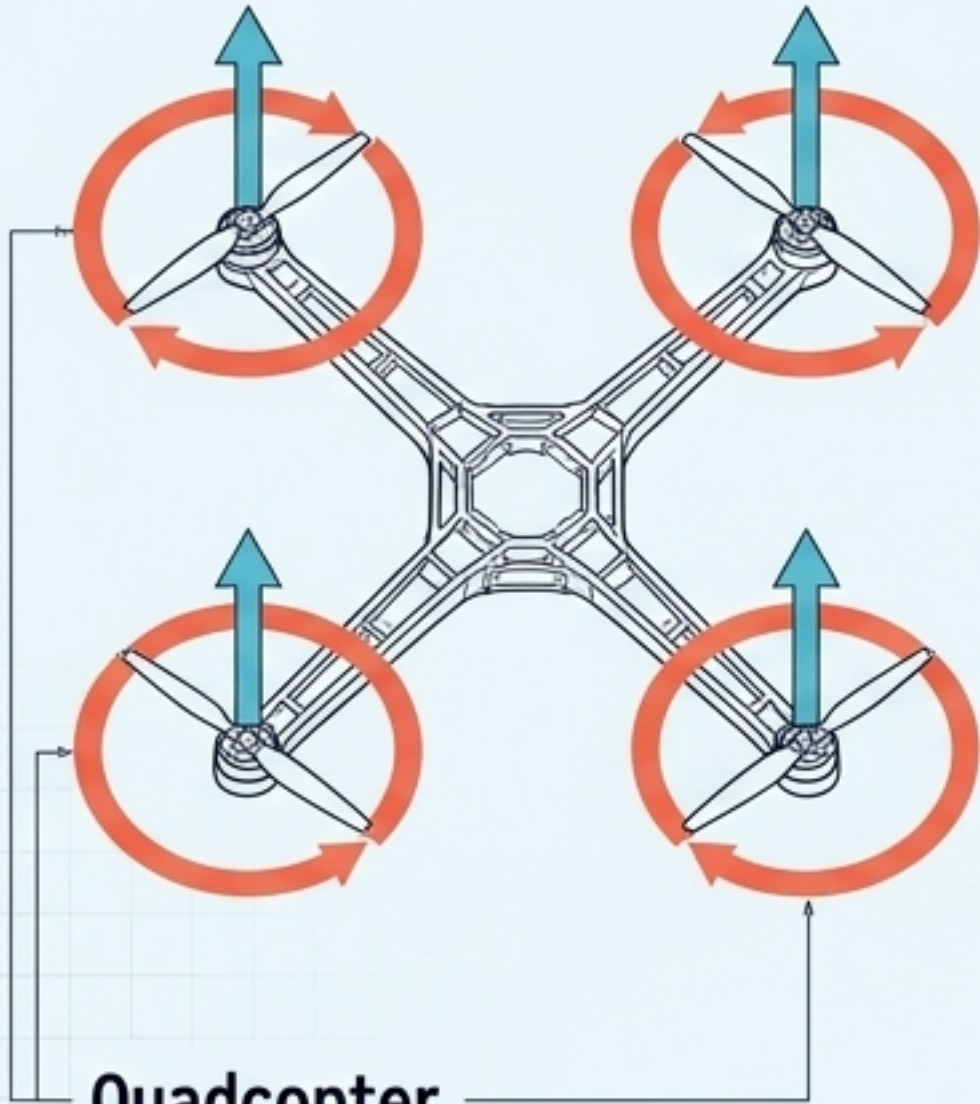
NATO vs. Commercial Size Classifications

	Classification	Size Metric	Performance Metric
1	Very Small	<9 kg	<0.12 km altitude
2	Small	0.51–2 m ²	<185 km/h
3	Medium	5–10 m ²	<200 kg payload
4	Large	>10 m ²	<600 kg payload

NATO Classes (GTOW & Service Ceiling)

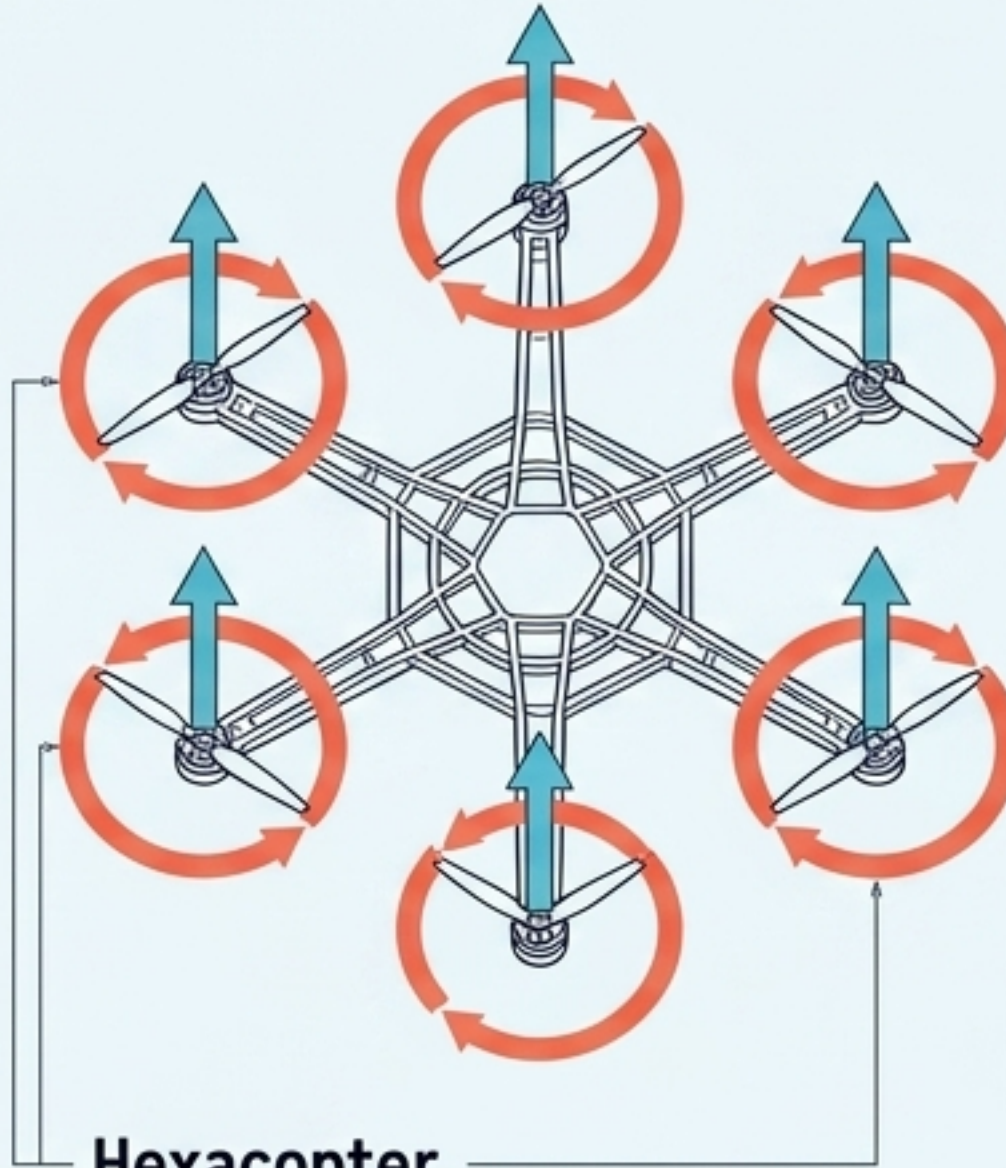
Class I:	Micro / Mini / Small
Class II:	Tactical
Class III:	MALE / HALE / Strike

Multi-Rotor Kinematics: Torque Cancellation and Systems Redundancy



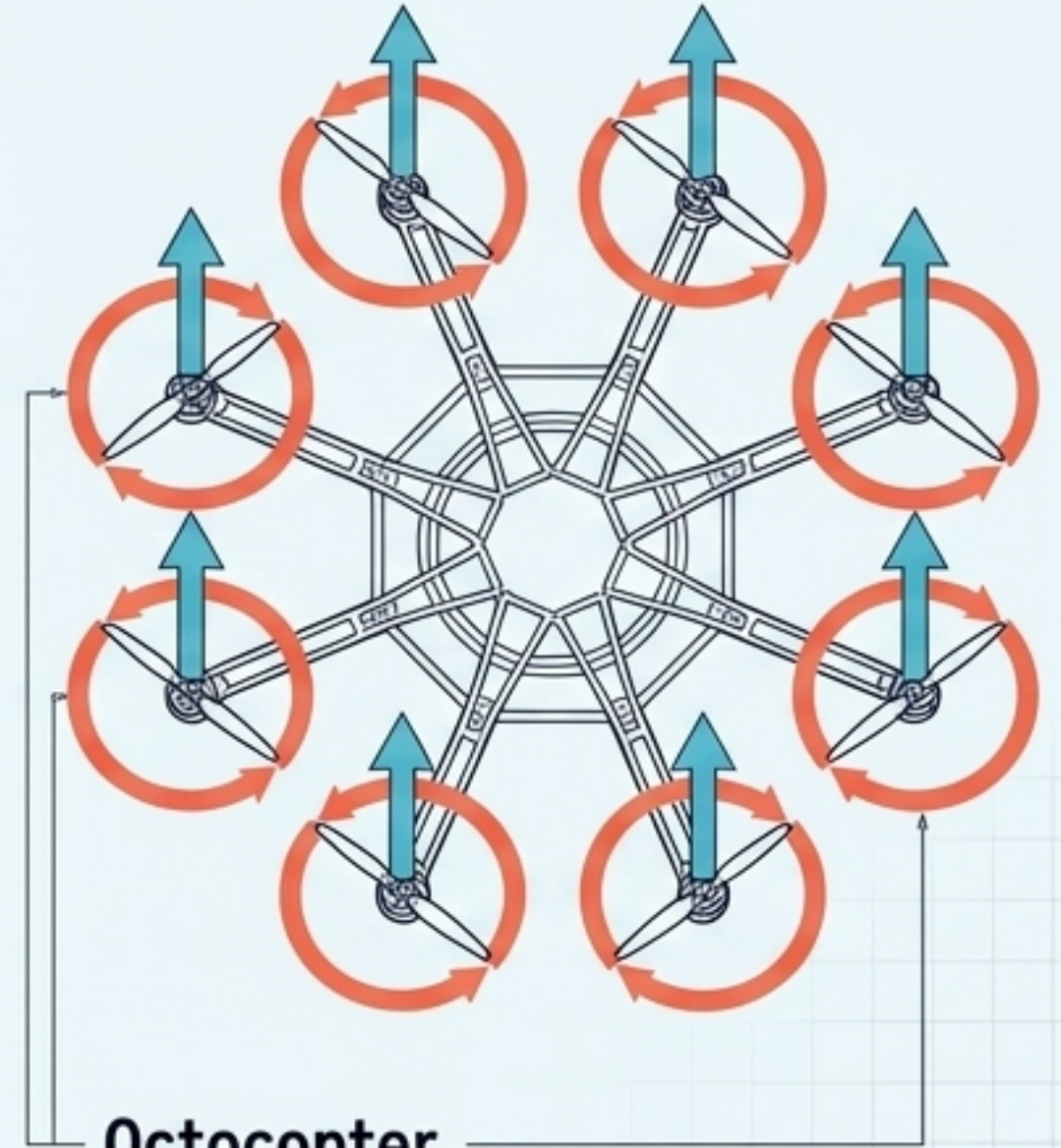
Quadcopter

- Zero redundancy. Symmetrical thrust.
- Catastrophic failure on single ESC/motor loss ('tumble').
- Differential thrust controls pitch/roll; torque imbalance controls yaw.



Hexacopter

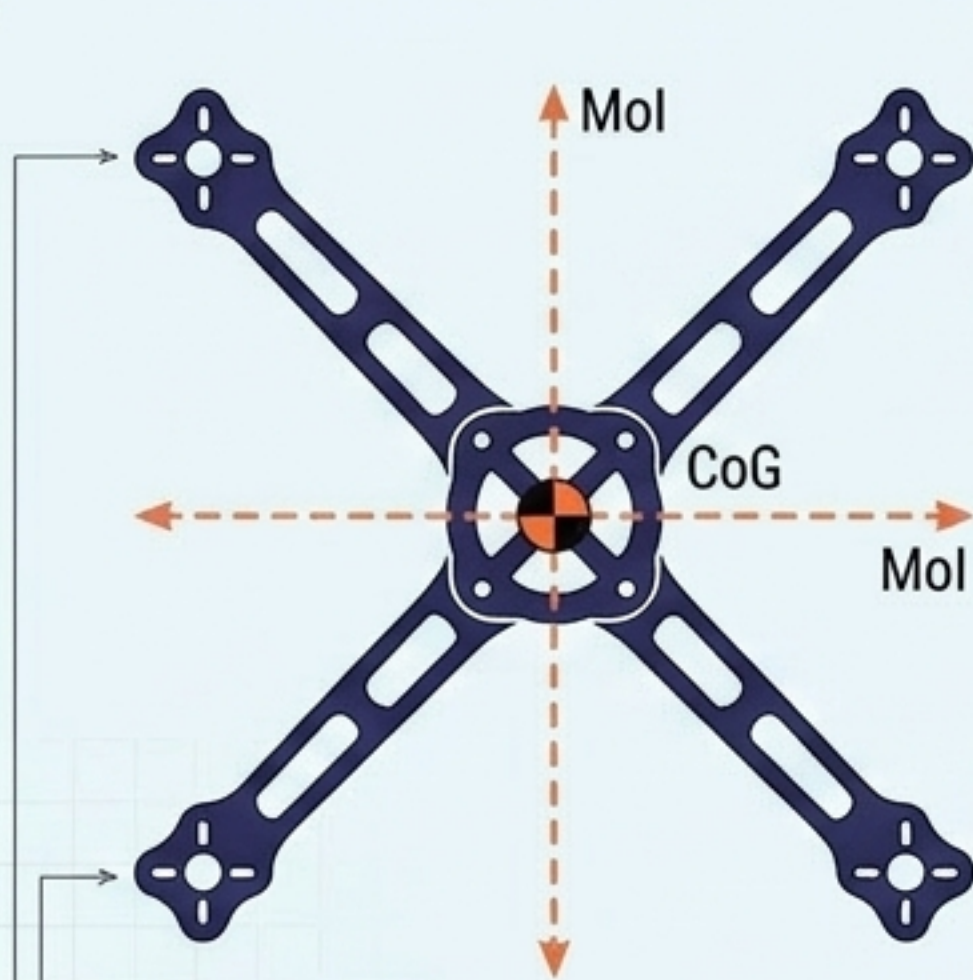
- N+1 Redundancy.
- Margin of safety allows sustained flight or emergency landing with single propulsion failure.



Octocopter

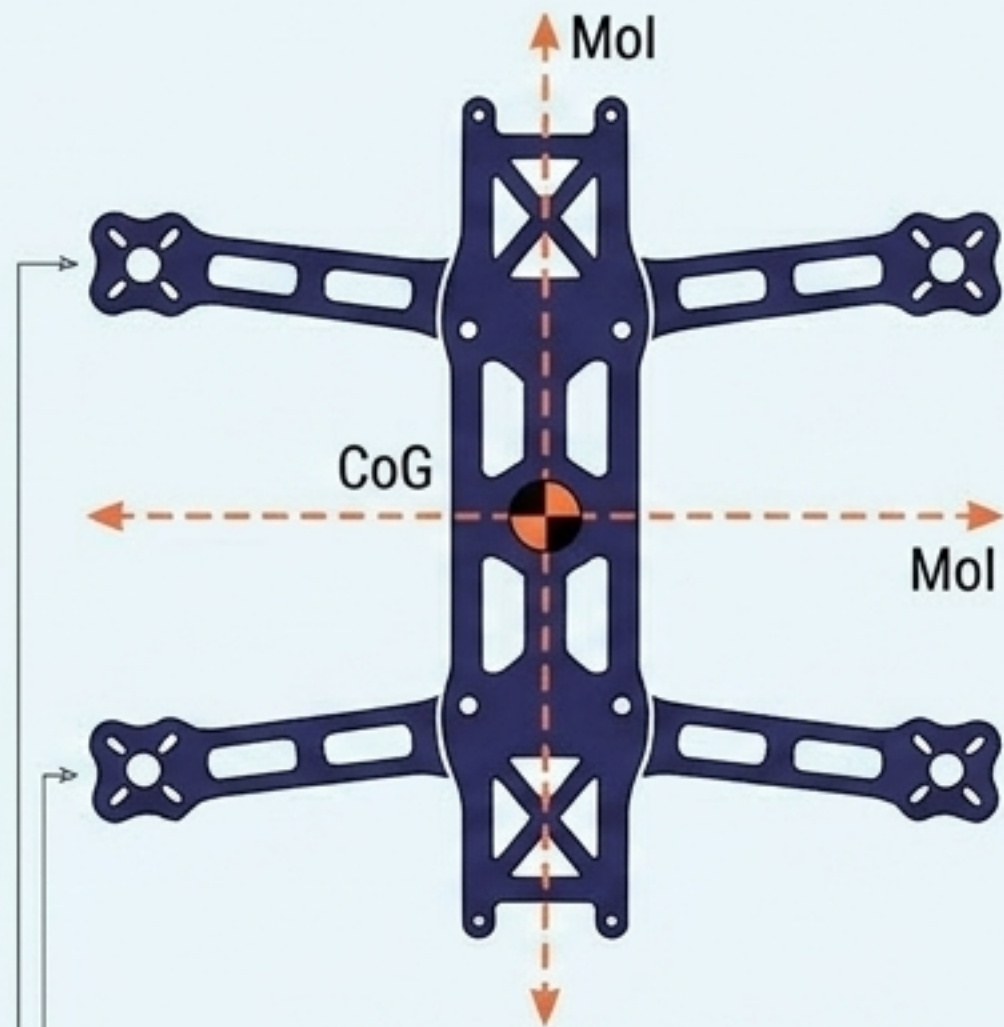
- N+2 Redundancy.
- Engineered for extreme stability and high-value payloads (LiDAR, cinema cameras).
- Maintains hover with multiple motor failures.

Frame Geometry Diagnostics: X, H, and Deadcat Structural Physics



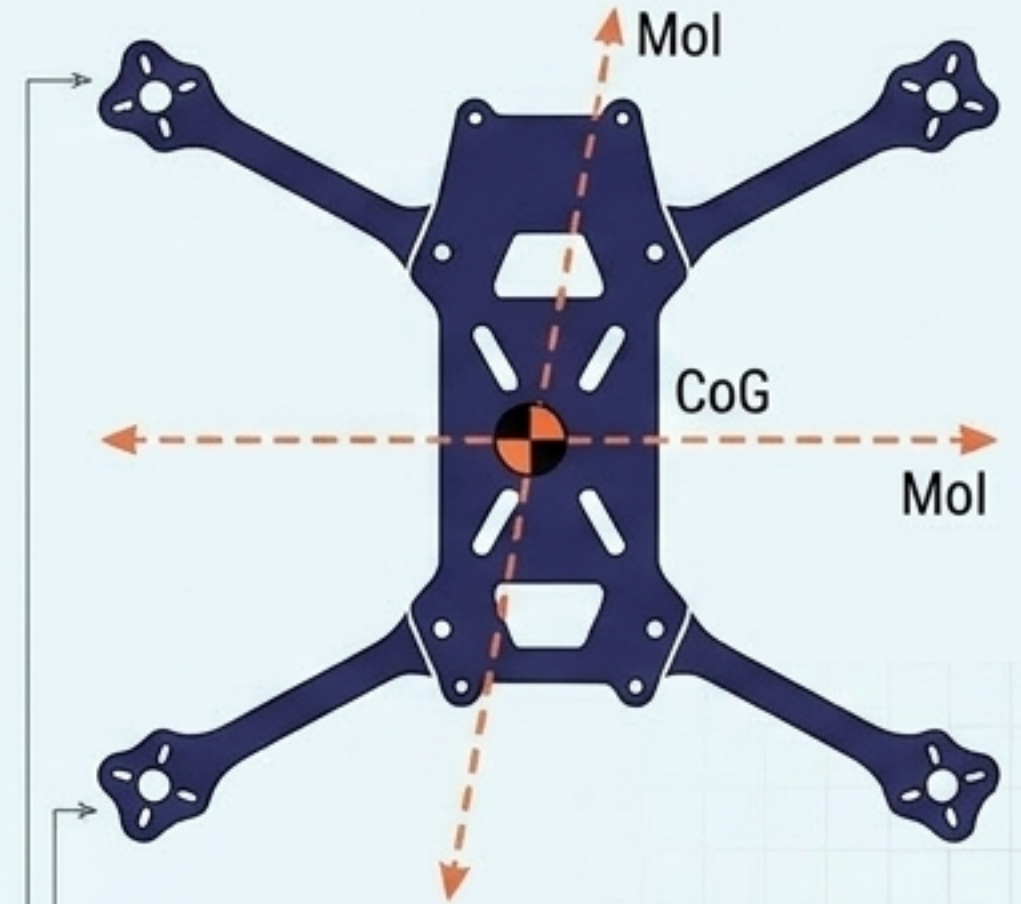
True X Frame

- Perfect symmetry.
- CoG perfectly aligned with thrust center.
- Lowest Mol on pitch/roll axes. Optimal for high-agility acrobatics.



H Frame

- Extended central fuselage.
- Higher pitch Mol.
- Allows for linear payload distribution but suffers from minor torsional flex under high loads.

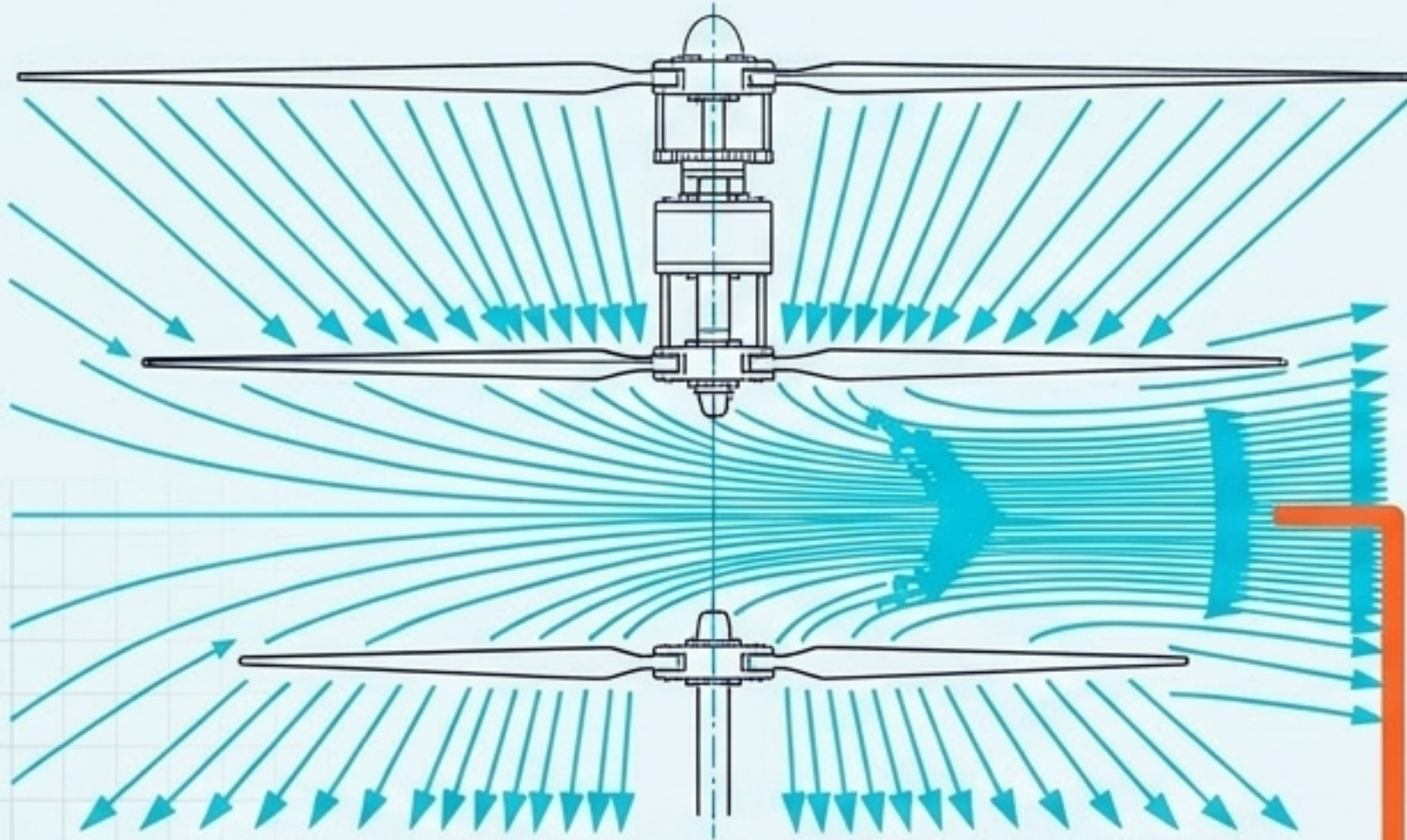


Deadcat Frame

- Front arms swept backward.
- Asymmetric geometry specifically engineered to clear the camera Field of View (FOV).
- Requires complex custom motor-mixing algorithms to compensate for off-center CoG.

Single-Rotor and Coaxial Dynamics: BEMT and Footprint Efficiency

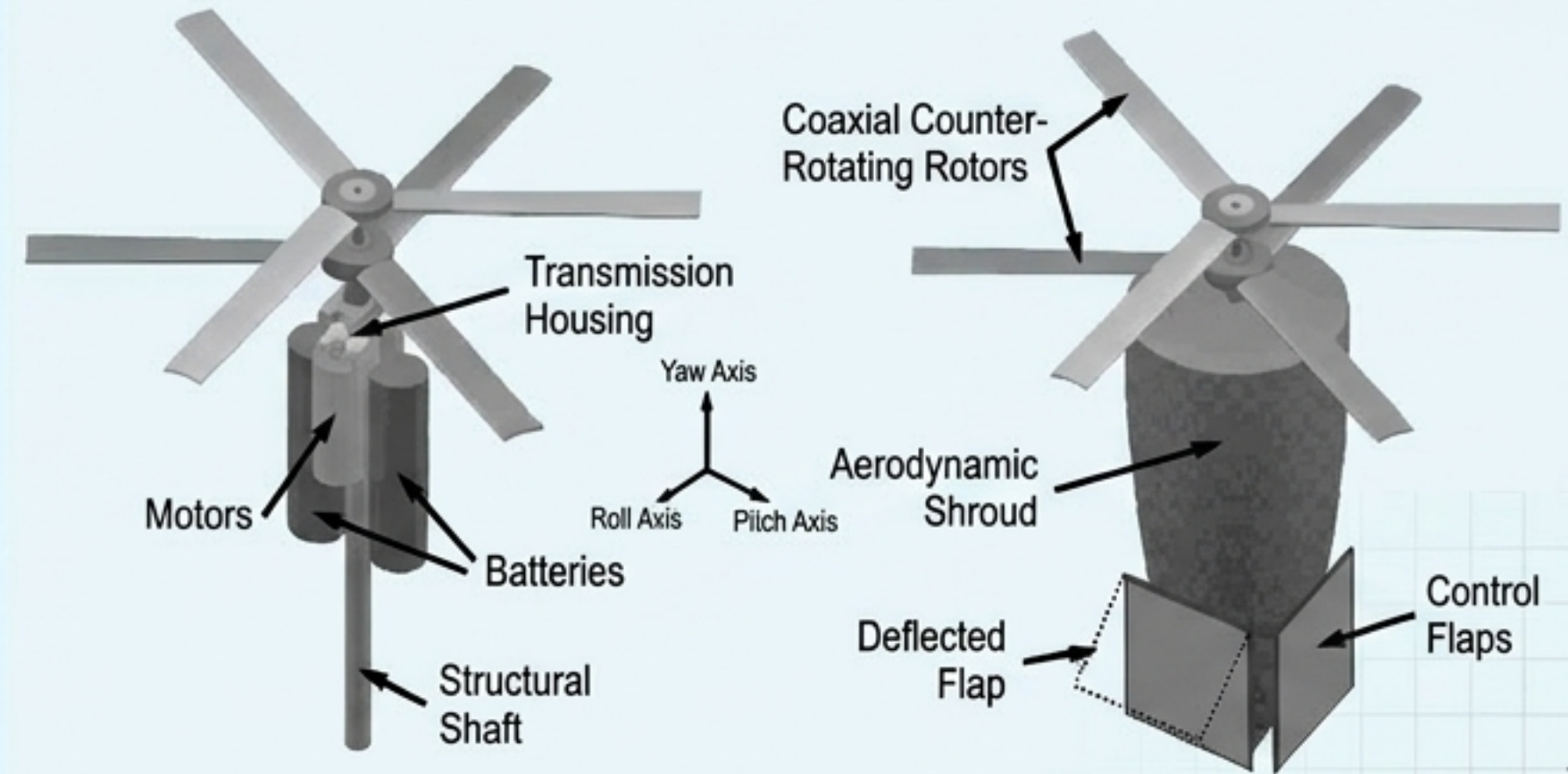
BEMT & Coaxial Downwash Visualization



1.28x Induced Power Penalty

Lower rotor operates in an effective climb velocity, requiring 28% more induced power than isolated rotors producing identical thrust.

Mechanical & Structural Schematic



Structural Advantages

- Total absence of tail boom and tail rotor.
- Drastic reduction in mechanical complexity.
- Superior footprint efficiency for confined naval or urban environments.

Fixed-Wing Aerodynamics: Lift Generation and Wing Loading

$$L = \frac{1}{2} \rho v^2 S C_L$$

Ailerons: Control
roll (tilt/turn)

Rudder: Control
Control yaw (steer)

Elevators: Control
pitch (climb/descend)

Wing Loading Panel



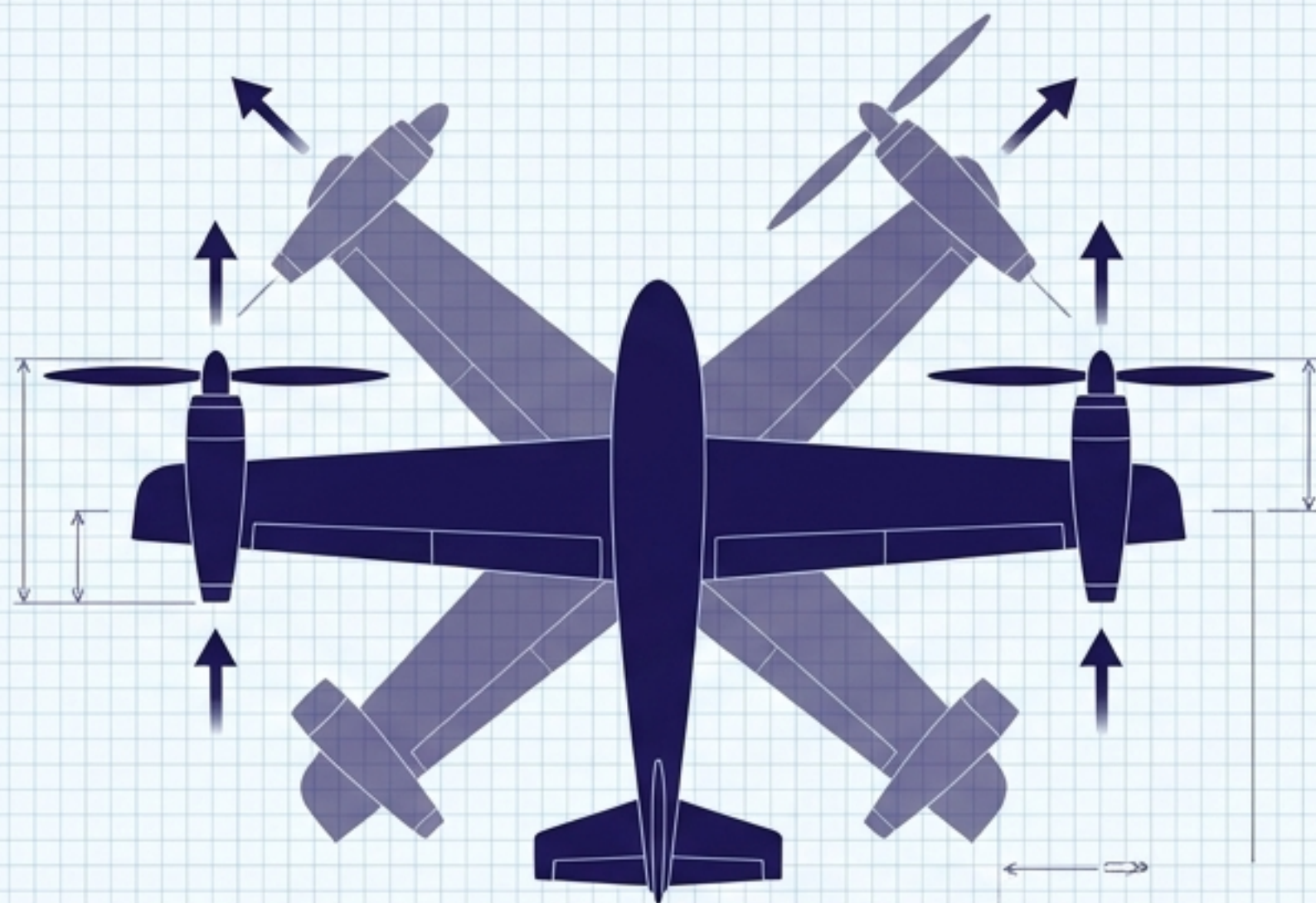
Lower W/S

Wing Loading (W/S)

- Ratio of Weight to Wing Area.
- Lower W/S (4-7 kg/m²) allows for slower takeoff speeds and hand-launches.
- Prioritizes extended endurance and low-speed loitering over high-speed cruise.

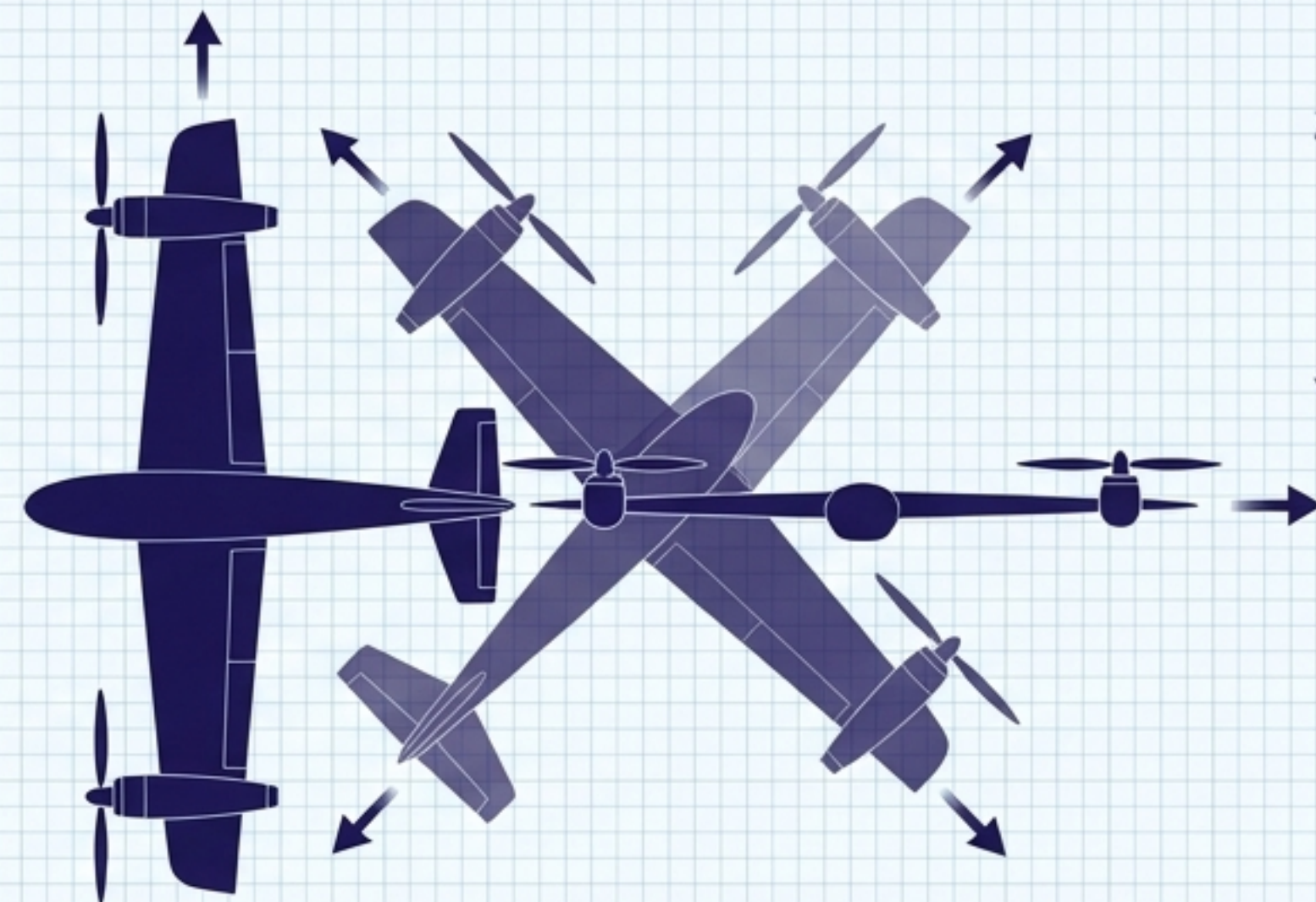


Hybrid VTOL Mechanics: Tilt-Rotor vs. Tilt-Wing Kinematics



Tilt-Rotor Telemetry

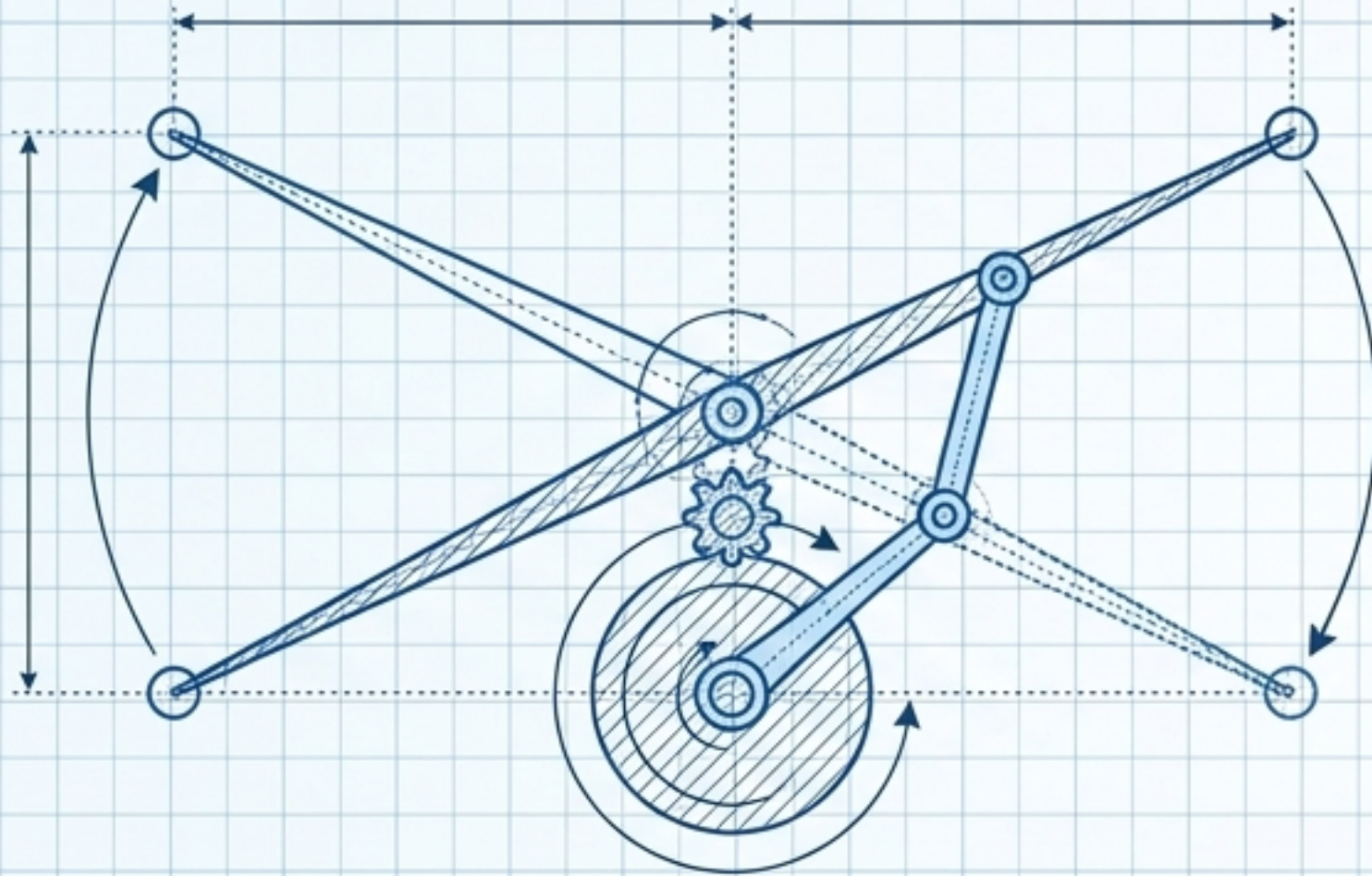
- Motors rotate 90°; main wing remains completely fixed.
- Higher mechanical complexity concentrated in nacelle pivot joints.



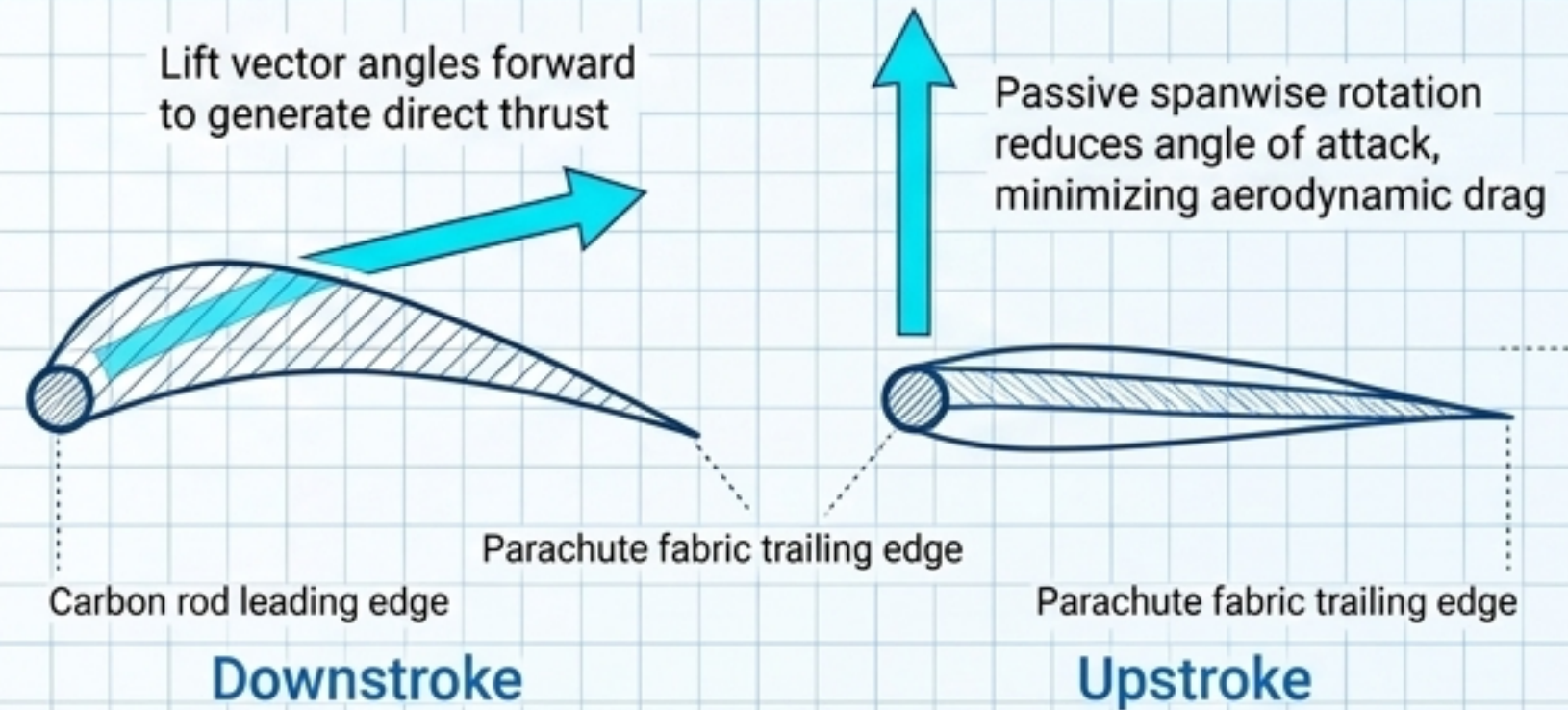
Tilt-Wing Telemetry

- Entire wing structure rotates.
- **Hover:** Pitch/roll via differential thrust. Yaw via flaperons in propeller slipstream.
- **Cruise:** Higher disc loading generates superior aerodynamic efficiency in forward flight.

Bio-Inspired Ornithopters: Flapping-Wing Kinematics and Passive Morphing



Flapping-Wing Kinematics



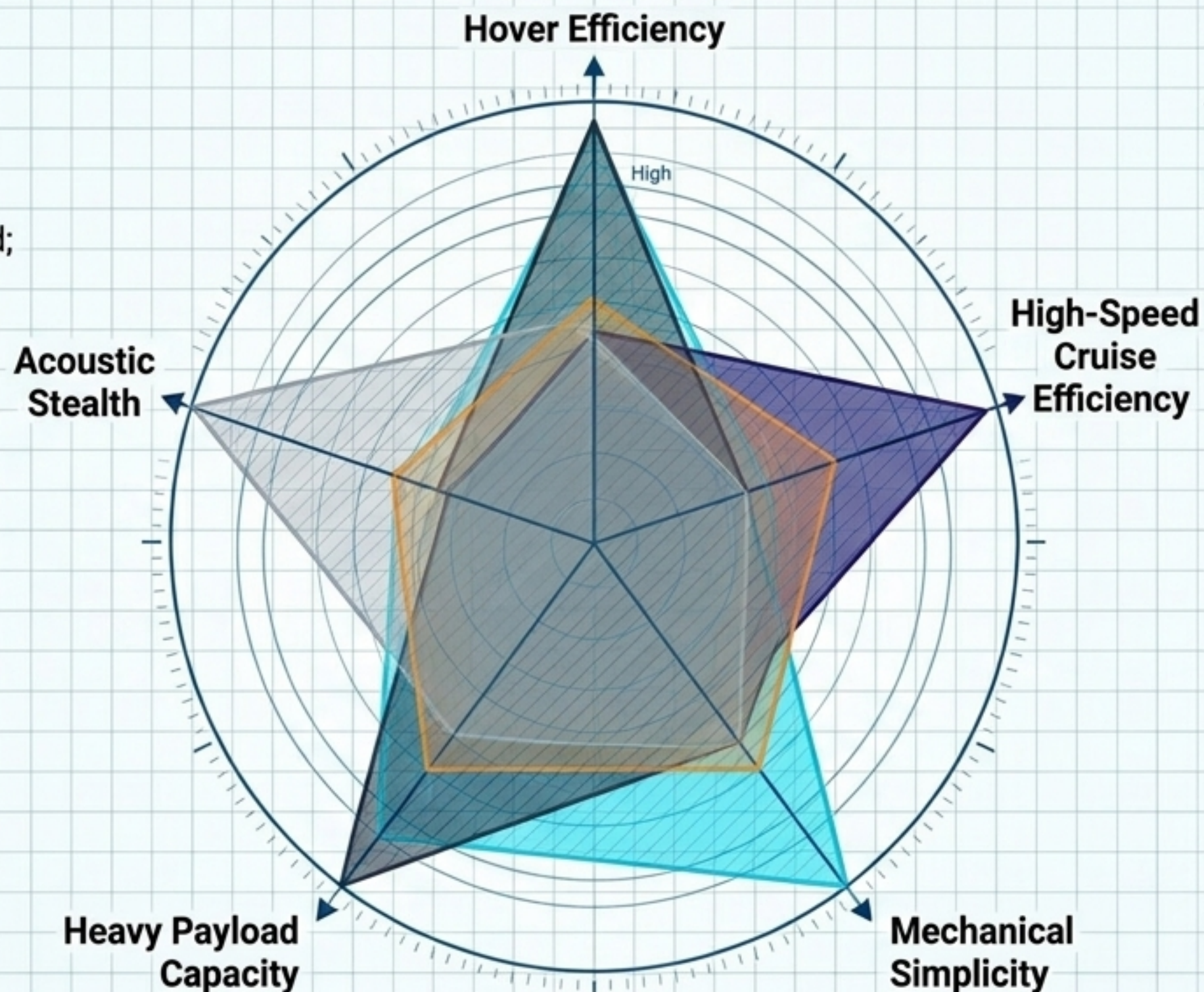
Aero-Elasticity Diagram

Operational Specs: Acoustic stealth | Extreme low-speed maneuverability | High mechanical resilience in cluttered environments



Payload vs. Endurance Synthesis: The Master Trade-Off Matrix

- Multi-Rotor**
(High Hover, Simple, Payload;
Low Cruise, Stealth)
- Fixed-Wing**
(High Cruise; Low Hover)
- Hybrid VTOL**
(Moderate across Hover,
Cruise, Payload)
- Coaxial**
(High Hover, Payload;
Low Stealth)
- Bio-Inspired**
(High Stealth; Low Payload)

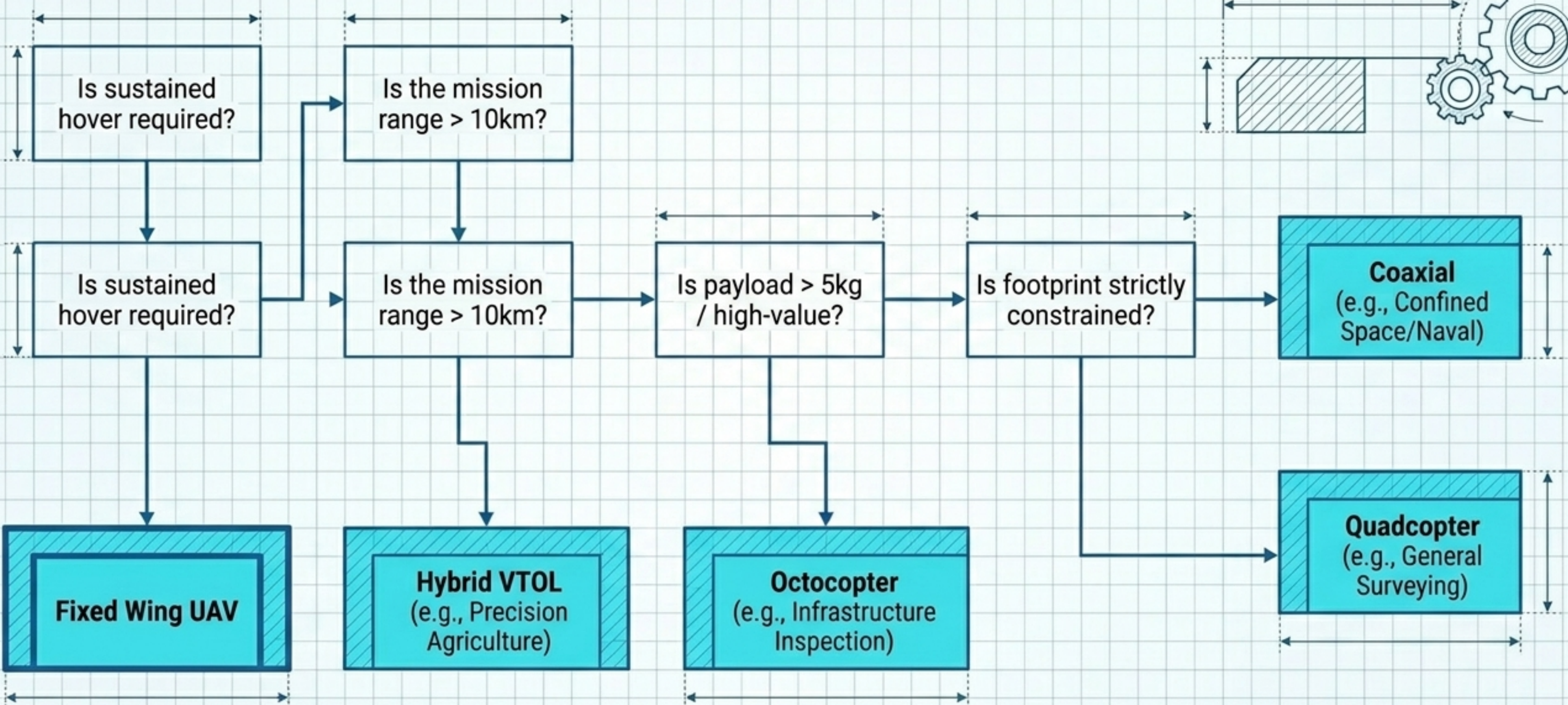


Absolute Rules of Physics

- Direct thrust requires exponential power scaling to increase payload.
- Aerodynamic lift scales linearly, offering vastly superior long-range efficiency.
- Mechanical complexity is inversely proportional to field reliability.



Operational Deployment: Algorithmic Selection Criteria





Future Configuration Vectors: Distributed Electric Propulsion (DEP)

